

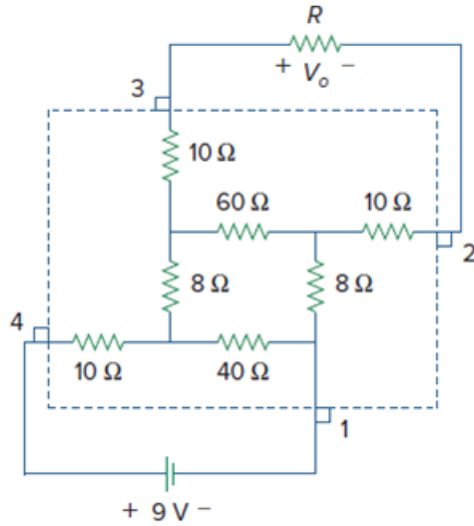
Problem

A resistance array is connected to a load resistor R and a 9-V battery as shown in Fig. 4.151.

(a) Find the value of R such that $V_o = 1.8$ V.

(b) Calculate the value of R that will draw the maximum current. What is the maximum current?

Figure 4.151:



Step-by-step solution

Step 1 of 12

Refer to Figure 4.151 in the textbook.

(a)

Draw the modified circuit diagram.

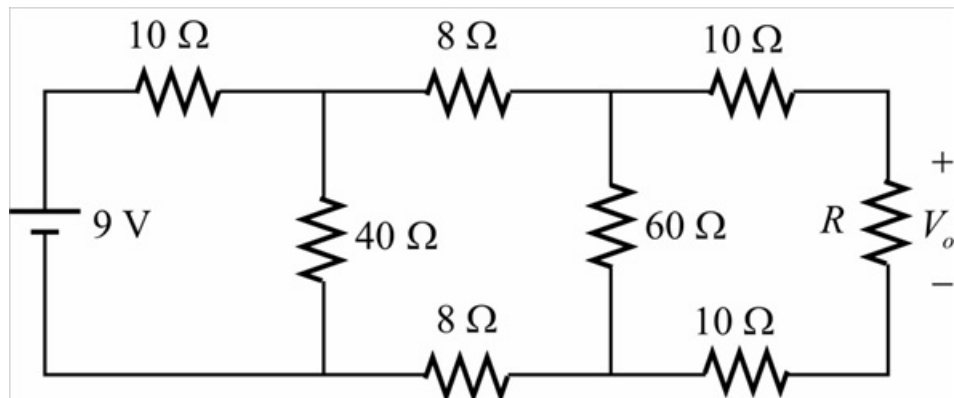


Figure 1

Step 2 of 12

Remove the resistance, R to calculate the Thevenin equivalent across the resistor, R .

Draw the circuit diagram to calculate the Thevenin's voltage.

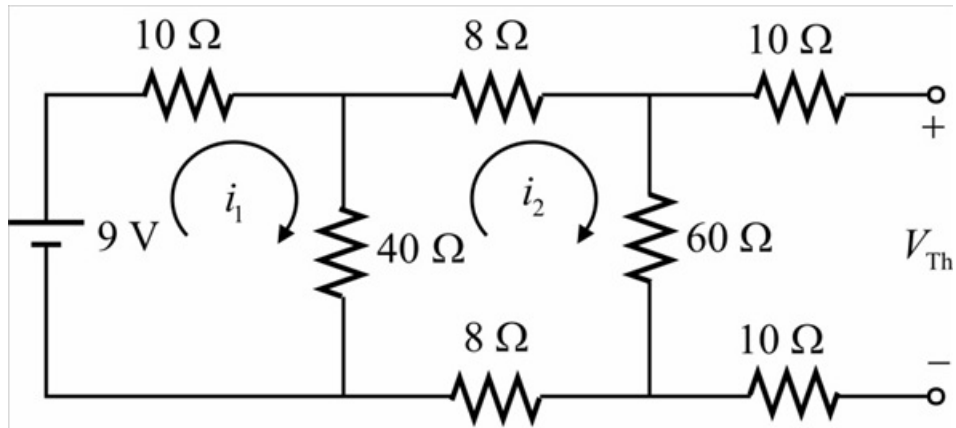


Figure 2

Step 3 of 12

Apply Kirchhoff's voltage law to the loop 2.

$$8i_2 + 60i_2 + 8i_2 + 40(i_2 - i_1) = 0$$

$$-40i_1 + 116i_2 = 0$$

$$i_1 = \frac{116}{40}i_2$$

$$i_1 = 2.9i_2$$

Apply Kirchhoff's voltage law to the loop 1.

$$-9 + 10i_1 + 40(i_1 - i_2) = 0$$

$$50i_1 - 40i_2 = 9 \dots\dots (1)$$

Substitute $2.9i_2$ for i_1 .

$$50(2.9i_2) - 40i_2 = 9$$

$$105i_2 = 9$$

$$i_2 = \frac{9}{105}$$

$$= 85.71 \times 10^{-3}$$

$$= 85.71 \text{ mA}$$

Step 4 of 12

Calculate the Thevenin's voltage.

$$V_{Th} = 60i_2$$

Substitute 85.71 for i_2 .

$$\begin{aligned} V_{Th} &= 60(85.71 \times 10^{-3}) \\ &= 5.143 \text{ V} \end{aligned}$$

Short circuit the voltage source to calculate the Thevenin,s resistance.

Draw the circuit to calculate the Thevenin,s resistance.

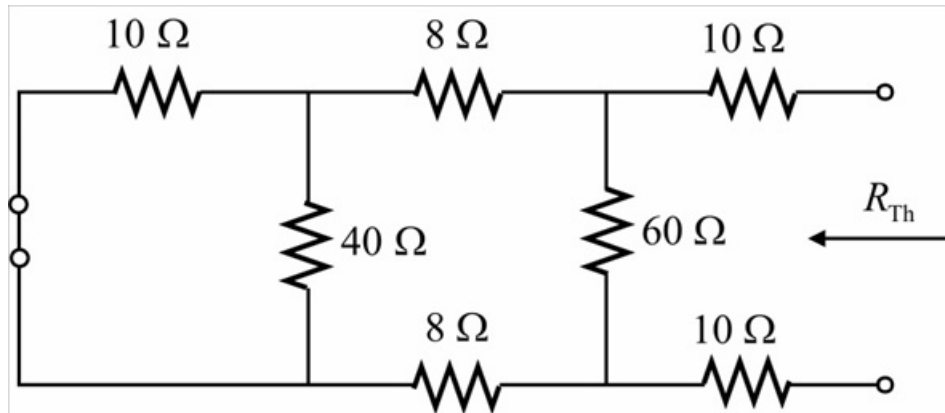


Figure 3

Step 5 of 12

From Figure 3, 10 Ω and 40 Ω are connected in parallel.

Calculate the equivalent resistance between the resistors.

$$\begin{aligned} R_{eq1} &= 10 \parallel 40 \\ &= \frac{10 \times 40}{10 + 40} \\ &= \frac{400}{50} \\ &= 8 \Omega \end{aligned}$$

Step 6 of 12

Draw the modified circuit diagram.

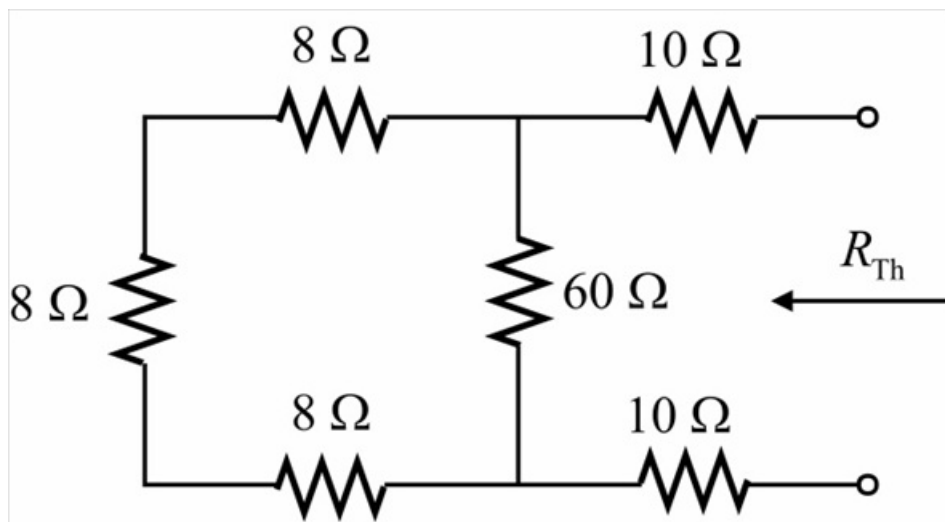


Figure 4

From Figure 4, three $8\ \Omega$ resistors are connected in series.

Calculate the equivalent resistance between the resistors.

$$\begin{aligned} R_{eq2} &= 8 + 8 + 8 \\ &= 24\ \Omega \end{aligned}$$

Step 7 of 12

Draw the modified circuit diagram.

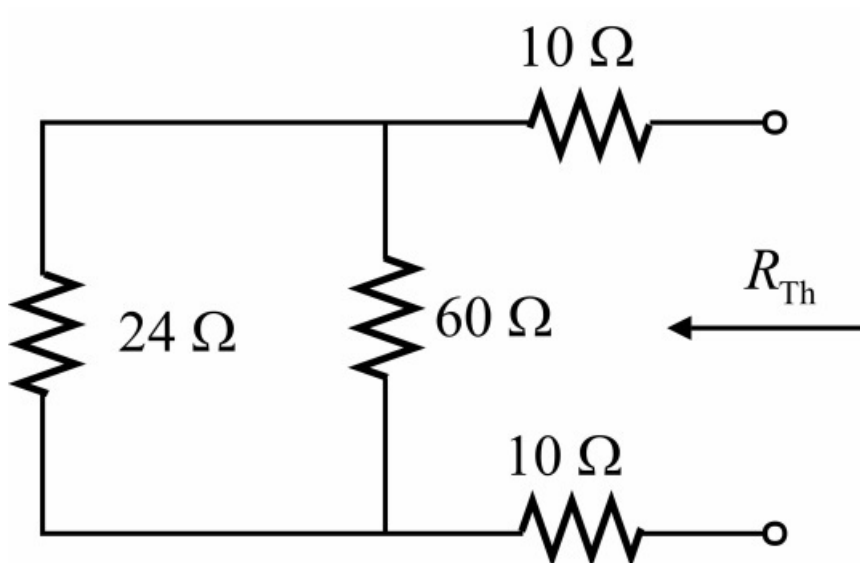


Figure 5

Step 8 of 12

From Figure 5, $24\ \Omega$ and $60\ \Omega$ are connected in parallel.

Calculate the equivalent resistance between the resistors.

$$\begin{aligned} R_{eq3} &= 24 \parallel 60 \\ &= \frac{24 \times 60}{24 + 60} \\ &= \frac{1440}{84} \\ &= 17.14\ \Omega \end{aligned}$$

Step 9 of 12

Draw the modified circuit diagram.

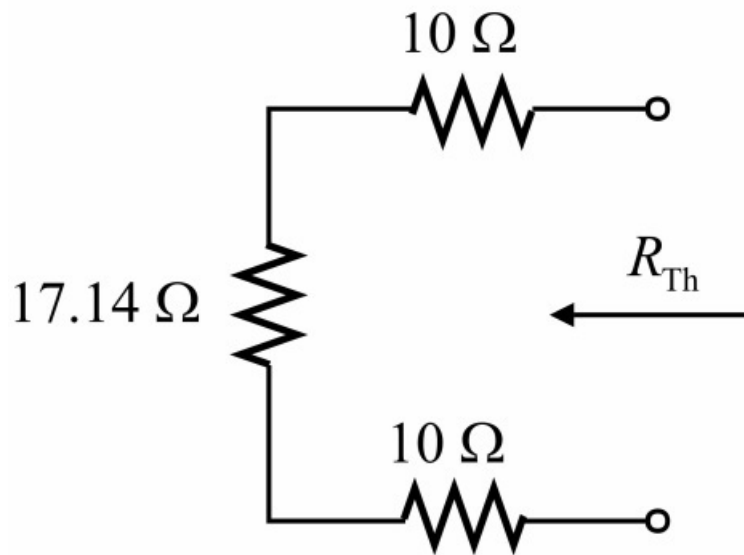


Figure 6

From Figure 4, two $8\ \Omega$ resistors and $17.14\ \Omega$ are connected in series.

Calculate the equivalent resistance between the resistors.

$$\begin{aligned} R_{Th} &= 10 + 10 + 17.14 \\ &= 37.14\ \Omega \end{aligned}$$

Step 10 of 12

Draw the equivalent circuit diagram for the circuit in Figure 1.

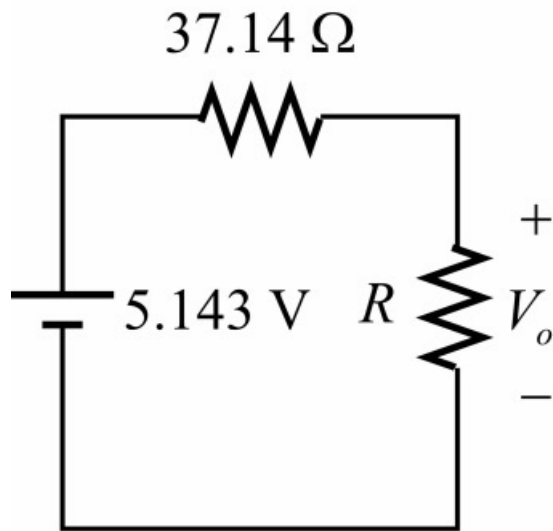


Figure 7

Step 11 of 12

Apply voltage division rule for the expression of output voltage in Figure 7.

$$V_o = \left(\frac{R}{R + 37.14} \right) (5.143)$$

Substitute 1.8 V for V_o .

$$1.8 = \left(\frac{R}{R + 37.14} \right) (5.143)$$

$$\frac{R}{R + 37.14} = 0.35$$

$$R = 0.35R + 13$$

$$(1 - 0.35)R = 13$$

$$R = \frac{13}{0.65}$$

$$= 20 \, \Omega$$

Thus, the value of the resistor, R is 20 Ω.

Step 12 of 12

(b)

The current through the resistance is maximum if the value of the resistance is low as possible.

The possible low value of the resistance is,

$$R = 0 \, \Omega$$

Thus, the value of the resistance for maximum current through it is $0 \, \Omega$.

Calculate the maximum current.

$$\begin{aligned} I &= \frac{V_{Th}}{R_{Th}} \\ &= \frac{5.143}{37.14} \\ &= 138.476 \times 10^{-3} \\ &= 138.476 \, \text{mA} \end{aligned}$$

Thus, the maximum value of the current is $138.476 \, \text{mA}$.